

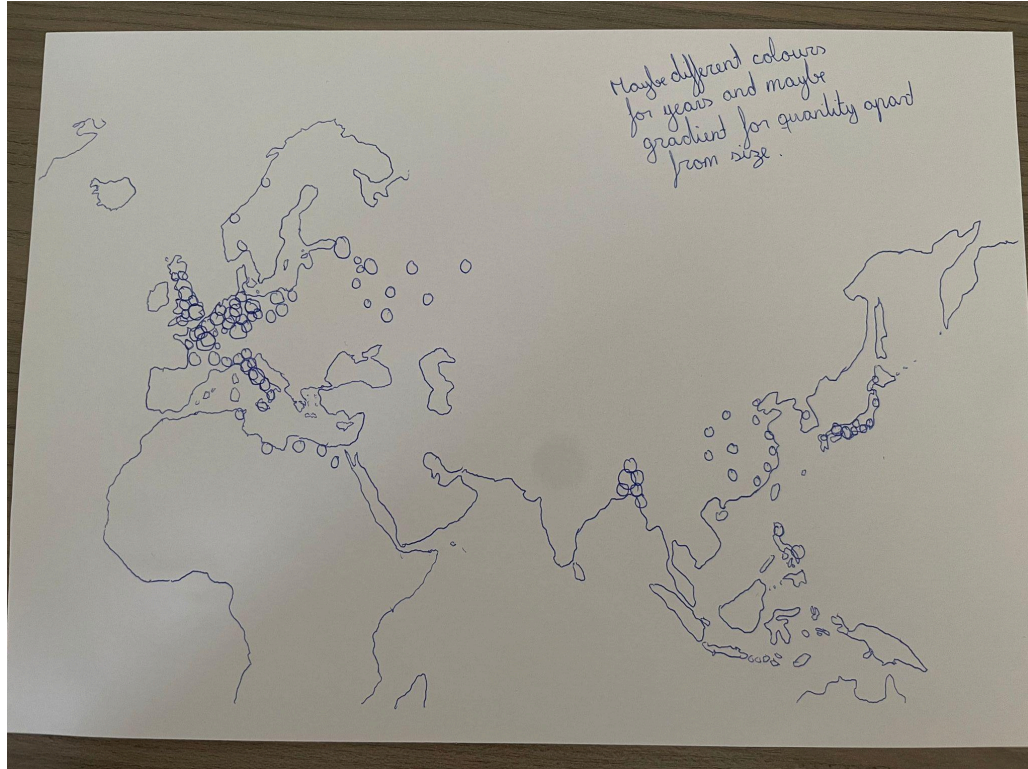
Process Book:

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Initial Visualisation

The first part of the project was to come up with a visualisation that would be worth implementing. We ran through a number of ideas, but a large part of it was ensuring it was something that we liked. We ended up choosing something to do with history, it was something we thought not too many people would do, and we thought was interesting. As a result, we scoured for databases that felt interesting, and then we came across the THOR database. Not all of it was publicly and freely accessible, but on Kaggle, we found 4 subsections of it, for 4 different major wars which is enough.

What we chose then was to do a sort of map where we would display what places were bombed and how many bombs were dropped. This was an idea brought about by many casualty maps, in many videos on history online, to display the casualties in war, it is displayed as a circle in the country with a larger circle being a larger casualty count. Alternatives include a list next to the map, and other such visualisations. We wanted to do something similar. Which is where the visualisation brought us, here is the sketch that we came up with, this was where we started:

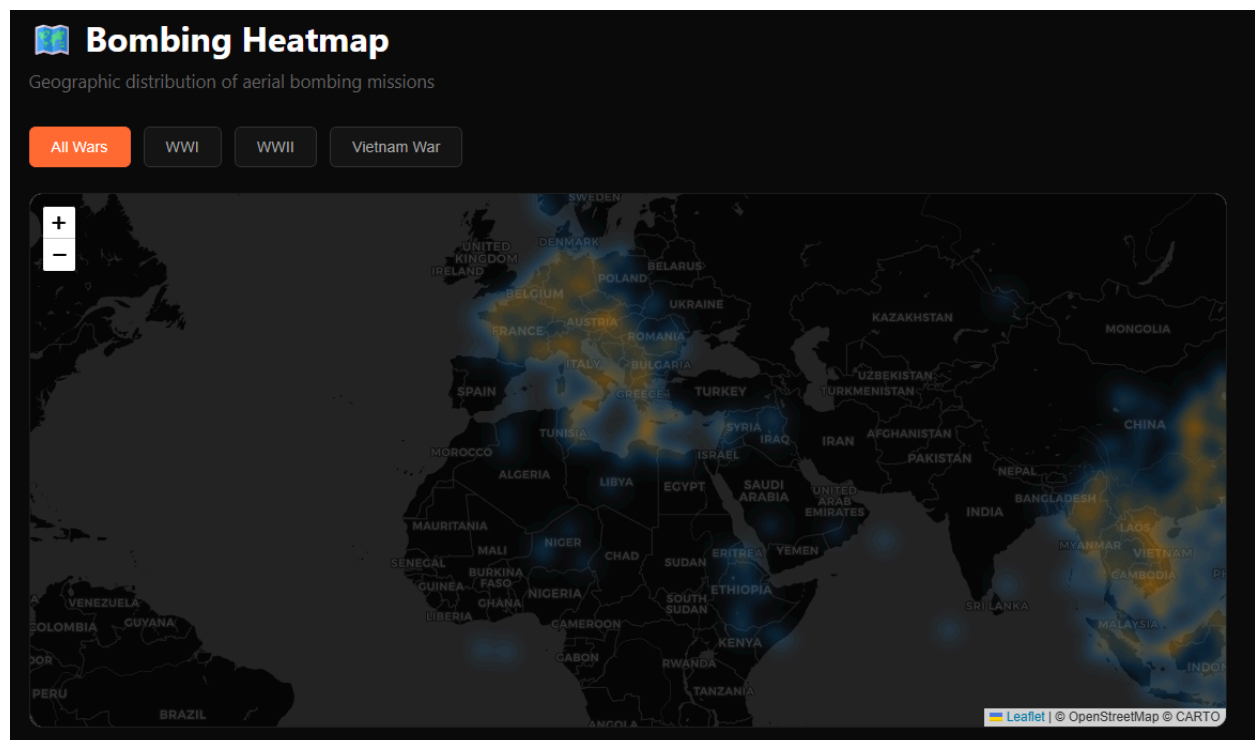


We also decided that the database was too large and needed pre-processing. We created two files, a processor, and a splitter. The processor would remove all entries that did not have all the necessary information, such as not having coordinates, tonnage, number of aircraft, etc... The splitter was charged with dividing the Vietnam THOR database into 10 pieces, as it was too large to upload to Github on its own.

Design Decisions & Visual Identity

At the beginning, we thought that our initial visualisation would be enough to tell the story. But when we started prototyping, we realized pretty quickly that a map alone doesn't really explain much. You see where the bombs fell, sure, but you have no idea about the timeline, who did it, or what planes were used. It's impressive visually but it lacks depth.

We also had to change from the initial idea to a heatmap, it is quite similar, and does the same thing, uses colour on a map to show where the bombs fell, and how many in one place. The problem with the initial idea, was that it was impossible to choose how to group the bomb locations into circles, what counted as the same place and what not, for this we ended up doing a heatmap, which is quite similar, but has a preestablished way of doing this. Here is the final heatmap result:



We then decided to build a multi-section page that combines the map with several charts (timeline, countries, aircraft, top targets). The map stays as the emotional centerpiece because it makes the scale of destruction immediately visible, but the charts around it give the analytical context that you need to actually understand the data. We think this combination works much better than either approach alone.

Why This Skeleton

We went with a single-page vertical scroll instead of a classic multi-tab dashboard. The reasons were:

1. It follows a narrative logic. You start with the hook, the big numbers, then go deeper into where, when, who, and what.
2. It's simple to use. Anyone can just scroll. There's no complex navigation to learn, no hidden panels to discover.
3. Instead of having separate pages for each war, we put filter buttons in each section. This way you can instantly compare — you see the same chart change between WWI and Vietnam, which makes the escalation much more concrete than just reading numbers.

The order of sections was thought carefully:

- Hero stats -> hook the user with big impressive numbers
- Map -> give spatial intuition
- Timeline -> show temporal evolution
- Countries & Aircraft -> explain who did it and with what
- Top Targets -> provide the most detailed view

Aesthetic Choices

For the visual identity we wanted something that feels somber, like the topic, but still engaging, hence the emojis used and other highlights.

- Dark background : sets a tone that fits the subject, which is dark. Also makes the colored data elements stand out more and is easier on the eyes for data-heavy pages.
- Orange accent : we picked this because it evokes fire and explosions, which is coherent with the bombing theme. It's used only for highlights (titles, active buttons, chart accents) so it doesn't change the tone to feel like it is too vibrant.
- War-specific colors (red=WWI, orange=WWII, green=Vietnam): allows the user to immediately recognize which conflict they're looking at, across all charts.

- CARTO Dark Matter map tiles: keeps the dark aesthetic while still showing country borders and city names. A light map would have broken the whole visual coherence.
- Fade-in animations on scroll: sections appear progressively as you scroll down, which reinforces the feeling of discovering the story step by step.

Challenges Faced

1. The Vietnam Problem (Scale Imbalance)

Vietnam has 3.5 million rows in the dataset. That's roughly 40 times more than WWII and 3000 times more than WWI. In any chart showing all wars together, Vietnam completely crushes everything else, the other wars become invisible. We dealt with this by aggregating data at a monthly level for the timeline, limiting bar charts to top-N entries, adding log scale options, and most importantly giving the user the ability to filter by war so they can look at smaller conflicts without Vietnam dominating.

2. Heatmap Radius vs Zoom Level

This one was surprisingly annoying. If you set a heat radius that looks nice at the global zoom, it becomes a giant unreadable blob when you zoom into a specific region. And if you optimize for close-up, the global view shows nothing. We ended up implementing a dynamic radius formula that adapts to the zoom level ($\text{radius} = \text{Math.max}(2, 25 - \text{zoom} * 1.5)$), which took quite a bit of trial and error to get right.

3. Browser Performance

Even after aggregating the data server-side, `map_data.json` still has thousands of points. On lower-end laptops we noticed lag when switching between wars, especially for Vietnam. We fixed this by pre-aggregating coordinates into grid cells during preprocessing (to reduce point count), and by fully clearing and rebuilding the heat layer on each filter change rather than trying to update it incrementally.

4. Chart.js vs D3.js

We debated a lot about whether to use D3.js for everything (since it's what the course emphasizes). In the end we went with Chart.js for the statistical charts because it's faster to get a polished result, built-in animations, responsive by default, and nice tooltips out of the box. D3 gives you more control but the development time is significantly higher. We felt it was more important to have a complete, working storytelling experience than to spend all our time on low-level SVG code for one perfect chart.

5. Synchronizing Filters Without a Framework

Each section uses a different library (Leaflet for the map, Chart.js for charts, vanilla JS for the table), but they all need to react to the same war filter. Without React or Vue to manage state, we had to be very disciplined: one shared `currentWar` variable and dedicated `update*()` functions called from every button's event listener. If anything gets out of sync (map showing Vietnam while chart shows WWII), the whole experience breaks.

6. Narrative vs Exploration

We constantly had to balance between making a guided story (fixed scroll order, editorial tone) and an exploratory dashboard (filters, user-driven discovery). Too much narrative and the user feels passive; too much freedom and there's no story anymore. Our compromise was: the scroll structure provides the narrative, but the per-section filters provide the freedom. The page suggests how to read it but never forces you.

Peer Assessment:

Here is what each person did: we divided the work into large parts and assigned them to individuals, we did not divide the work into smaller parts, this is due to the fact that often, people were busy and only one or two of us were free, making sure everybody did something at some point.

- The initial visualisation and the initial data processing was idealised and done by Jaime Oliver Pastor.
- The Initial Aesthetic choices and skeleton structure of the website was done by Antony Picard.
- The further processing of data into a json file that we can use using a python file was done by Alexandre Majchrzak.
- The implementation of the widgets using JavaScript was done by everybody, Alexandre Majchrzak and Jaime Oliver Pastor worked on the main map widget and Antony Picard did the rest.
- Lastly, the final polishing and rechecking with correcting a few errors was done by Jaime Oliver Pastor and Alexandre Majchrzak.
- The screencast was filmed by Alexandre Majchrzak.